

GSTAM BOOK

Behind it: a domain, a challenge, and
a path forward

SUMMARY

1 | Overview Of CSTAM Book

First Domain : eSports & Gaming Infrastructure | 2

3 | Second Domain : Internet of Things & Industry 4.0

Third Domain : Cloud Computing & Infrastructure | 4

5 | Fourth Domain : HealthTech

Automation | 6

7 | General Guidelines

Contact us | 8

Introduction to the CSTAM Book

Welcome to the CSTAM Book !

The **CSTAM Book** is the official guide to the technical challenges of the **Computer Society Tunisian Annual Meeting (CSTAM 3.0)**.

Throughout CSTAM Book, participants will have the opportunity to work on projects proposed by **IT companies**, addressing real technical challenges in their companies or innovative solutions.

Our Vision

The framework of Technical Challenges introduced in the previous editions has not disappeared, it has evolved.

Instead of offering a single challenge, CSTAM Book presents a complete ecosystem of technical challenges across multiple IT domains.

CSTAM 3.0 is designed to be **Tunisia's Bridge to the Global Tech World**, an immersive platform where education, innovation, competition, and industry come together.

CSTAM Book Framework

The CSTAM Book is organized around two strategic technology tracks that represent two of the most transformative forces shaping modern industries:

Automation

Cybersecurity

Depending on the challenge, these tracks may operate independently or together, reflecting the realities of today's technological ecosystems.

Challenges Domains

The CSTAM Book brings together technical challenges spanning the following technology domains:

eSports &
Gaming Infrastructure

Internet of Things
(IoT) & Industry 4.0

HealthTech

Cloud Computing
& Infrastructure

Automation

The CSTAM Book is your guide to discover new technologies, solve problems of IT companies and meet with experts who will guide you along the way.

Call for action, take the plunge, and join us in creating innovative solutions!

eSports & Gaming Infrastructure



Challenge Title: Intelligent eSports Venue Management Platform

Challenge Reference: CSTAM-NINETY

Company Collaborator: Ninety Gaming House

Assigned Mentor:

Mr. Aymen Sankouhi – Ninety Gaming House Technical Leader

1. Company Introduction & Challenge Context

Ninety Gaming House operates in the premium eSports sector, providing high-performance competitive gaming environments. To support scaling operations, the company seeks to modernize its software stack through venue automation, local network orchestration, and self-hosted infrastructure. The goal is to optimize the daily management of gaming stations and improve the player experience without relying on external dependencies.

2. Problem Statement

The core challenge is to build a centralized, lightweight software application to remotely control multiple gaming client PCs from a host computer over a local IP network. The system must securely manage workstation access, monitor real-time status, handle session timers, and execute remote commands all without consuming heavy background resources on the high-end gaming rigs.

3. Technical Mission

Challenge Objective:

Architect and deploy a secure, low-latency, and resource-efficient local network orchestration platform that empowers venue administrators to seamlessly control, monetize, and monitor high-end eSports client machines from a centralized dashboard.

Teams must build a system comprising a Frontend, Backend, Authentication layer, and custom Desktop Agent. The project will be evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Secure Access & Roles [6 Points]:** Robust authentication, login, and permission management.
- **Node Tracking [5 Points]:** Dynamic PC network registration and real-time online/offline status monitoring.
- **Session & Financial Control [7 Points]:** Automated start/end of gaming sessions synchronized with billing.
- **Remote Administration [6 Points]:** Instant remote screen locking and shutdown commands from the admin dashboard.
- **Hardware Telemetry & Anti-Theft Alerts [5 Points]:** Real-time monitoring of client GPU/CPU temperatures, fan speeds, and peripheral connection status (USB keyboard/mouse disconnect alerts).

- **Electronic Wallet System [5 Points]:** Integrated wallet management for user balances and quick transactions.
- **Subscription & Membership Plan Management [4 Points]:** Tiered memberships, recurring access plans, and gamer profiles.
- **Advance Reservation System [4 Points]:** Remote and local booking system for gaming stations.
- **Multi-Agency Support [4 Points]:** Multi-location and multi-branch management capabilities from a single master dashboard.
- **Centralized Catalog of Available Games [4 Points]:** Unified library management and remote game launching.

Required Deliverables :

- Source code repository (Backend, Frontend, and Desktop Agent).
 - Basic API Documentation (e.g., Postman collection or Swagger).
 - A short video demonstration showing the MVP features in action.

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Client background resource footprint [15 Points]:** Ensuring the desktop agent uses minimal CPU/RAM so gaming performance is unaffected.
- **Network synchronization and resilience [10 Points]:** Handling network drops and syncing states seamlessly over the local IP network.
- **Security and Operating System lockdown [5 Points]:** Preventing unauthorized bypass of the client lock screen.

Required Deliverables :

- Performance benchmark report (detailing CPU/RAM consumption of the client agent).
- Network architecture and communication flow diagram.
- Security strategy document detailing OS lockdown measures.

Phase 3: UI/UX & Documentation

Total score: 10 Points

Refine the project for final evaluation by improving usability and preparing all required deliverables.

- **Quality of UI/UX [10 Points]:** Intuitive Admin Dashboard for staff and a clean, unobtrusive Client UI for gamers.

Required Deliverables :

- Finalized UI/UX design and user experience improvements.
- Technical report documenting the project, implementation, and results.
- A demonstration video showcasing the project's features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch
- A live demonstration
- Q&A session with the jury members

Optional Advanced Features

Bonus Points

- **Dedicated Mobile Application for Clients [+2 Points]:** Mobile portal for remote station booking, wallet top-ups, and live station availability checks.
- **LAN Tournament & Bracket Engine [+2 Points]:** Built-in tournament setup, bracket generator, and automated seat assignment.

4. Recommended Skills

While you are highly encouraged to innovate and bring your own ideas to the table, your team will have a strong head start if you are familiar with:

- **Client-Side Desktop Development:** C#, C++, or similar frameworks.
- **Server-Side & Backend:** Python, Go, Node.js, or similar languages.
- **Networking & Sync Protocols:** WebSockets, gRPC, or equivalent.
- **Infrastructure:** Databases, Caching, DevOps, and Deployment.

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Summer internships and PFE 2027 opportunities at Ninety Gaming House.
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all challenges**



IoT & Industry 4.0



Challenge Title: Autonomous Service Robot for Indoor Delivery

Challenge Reference: CSTAM-TUNIBOT

Company Collaborator: TuniBot

Assigned Mentor:

Mr. Skander Abdelhedi – TuniBot
Automation Engineer

1. Company Introduction & Challenge Context

Tunibot operates in the robotics and automation sector, specializing in the development of intelligent robotic solutions for real-world applications. **This challenge is entirely focused on software development.** Teams are not required to build hardware, the objective is to architect, simulate, and program an autonomous service robot using industry-standard open-source platforms.

2. Problem Statement

Current service operations require continuous human intervention for repetitive delivery tasks. The challenge is to develop a software solution that enables a simulated robot to autonomously understand and navigate an indoor environment, receive user delivery requests for food and beverages, move to predefined locations to execute the delivery, and automatically return to a docking station for charging when idle or low on battery.

3. Technical Mission

Challenge Objective:

To design, prototype, and demonstrate a complete software stack (Simulation, SLAM, Navigation, Task Management, and UI) that successfully orchestrates an autonomous indoor delivery robot in a virtual environment without human intervention.

Teams must build a system comprising a Simulation Environment, Robot Core Logic, and a User/Admin Interface.

The project will be evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Mapping & Localization (SLAM) [15 Points]:** Ability of the robot to accurately map an unknown indoor environment and locate itself within it.
- **Autonomous Navigation [15 Points]:** Safely planning paths and moving to predefined target locations without collisions.
- **Delivery Task Management [10 Points]:** System logic to receive a food/beverage request and process the delivery queue.
- **Auto-Docking Behavior [10 Points]:** Logic to trigger and execute a return-to-dock sequence.

Required Deliverables :

- Source code repository (ROS 2 workspaces, nodes, and UI backend).

- High-level Software Architecture Diagram.
- A short video demonstrating basic movement and mapping in the simulation.

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Dynamic Obstacle Avoidance [10 Points]:** Reacting safely to moving obstacles (e.g., humans walking) in real-time.
- **Complex Layout Handling [10 Points]:** Navigating through tight corridors, doorways, and recovering from navigation failures.
- **Battery & State Simulation [10 Points]:** Accurate simulation of battery drain and triggering the return-to-dock behavior before the battery dies.

Required Deliverables :

- Complete simulation environment package (Gazebo).
- Test case documentation detailing how edge cases (obstacles, low battery) were handled.
- Screen-recorded demonstration of the robot handling dynamic obstacles.

Phase 3: UI/UX & Documentation

Total score: 10 points

- **Quality of Interface [10 Points]:** An intuitive User Interface (web or mobile) allowing users to easily request deliveries, and an Admin view to monitor the robot's status and battery.

Required Deliverables :

- Finalized UI/UX design and user experience improvements.
- Technical report documenting the project, implementation, and results.
- A demonstration video showcasing the project's features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch
- A live demonstration
- Q&A session with the jury members

Optional Advanced Features

Bonus Points

- Multi-robot fleet coordination simulation **[+2 Points]**
- Voice command integration for placing orders **[+2 Points]**
- Visual recognition (simulating computer vision to detect a specific VIP client) **[+2 Points]**

- Simulated elevator integration for multi-floor delivery **[+2 Points]**
- Predictive maintenance and analytics dashboard **[+2 Points]**

4. Recommended Skills

While you are highly encouraged to innovate and bring your own ideas to the table, your team will have a strong head start if you are familiar with:

- **Robotics Software & Frameworks:**
ROS 2, Gazebo, Navigation2 (Nav2), SLAM toolboxes.
- **Programming Languages:**
Python, C++.
- **Protocols & Integrations:**
ROS communication, REST APIs, MQTT or WebSockets for the UI.

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Summer internships and PFE 2027 opportunities at TuniBot
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all Challenges**

Cloud Computing & Infrastructure



Challenge Title: The Resilient IP Optimizer

Challenge Reference: CSTAM-FELCLOUD

Company Collaborator: FelCloud

Assigned Mentor:

– Mr. Mohamed Mongi Benyaiche – CEO
FelCloud

– Mr. Oussama Zied – FelCloud Cloud &
Devops Engineer

1. Company Introduction & Challenge Context

FelCloud operates in the Cloud Hosting sector, specializing in virtualized environments and developer sandbox automation. To drive digital transformation, we continuously explore innovations in Automated Edge Routing, High Availability Gateways, and Dynamic IP Management. In today's rapidly evolving landscape, where dynamic cloud scaling is increasingly critical, we currently rely on static subnets and manual reverse proxies. As operations scale during high-density hackathons, we are actively seeking innovative approaches to eliminate single points of failure and automate dynamic IP recycling.

2. Problem Statement

Our main challenge is IPv4 address scarcity and gateway single points of failure. When multiple teams deploy sandboxes simultaneously or a gateway configuration fails, our systems experience network downtime and IP conflicts. We require a robust solution capable of automatically provisioning sandboxes, routing subdomains across redundant HA gateways with rollback capabilities, and aggressively recycling private IPs without disrupting existing infrastructure.

3. Technical Mission

Challenge Objective:

Participating teams must build a system evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Automated OpenStack Sandbox Provisioning [15 Points]:** API-triggered auto-provisioning of isolated student VMs/containers on OpenStack with dynamically assigned private IPs.
- **Dual Redundant HA Gateways [15 Points]:** Active/Standby or Active/Active edge gateways (Keepalived/VRRP/Octavia/HAProxy) routing incoming subdomains (e.g., team5.cstam.felcloud.tn) to private sandboxes with instant, seamless failover.
- **Automated IPAM & Reclamation Engine [10 Points]:** Managed OpenStack IP pool allocation with short leases and immediate release upon sandbox teardown or expiration.
- **Zero-Downtime Reload & Auto-Rollback [10 Points]:** Hot-reloading of routing rules with automated syntax validation and sub-second auto-rollback upon corrupt or invalid configurations.

Required Deliverables :

- Source code repository
- System Architecture & OpenStack HA Gateway Data Flow Diagram

- Short video demonstration showcasing active subdomain routing, dynamic hot-reloading, and simulated primary gateway failover

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Concurrent Provisioning Stress Test [15 Points]:** Rapid, error-free sandbox provisioning, IP allocation, and routing updates for 10+ simultaneous team registrations.
- **Automated Teardown & Lifecycle Security [10 Points]:** Automated cleanup pipeline purging expired sandboxes, proxy/DNS rules, and sanitizing IP allocations.
- **Network Isolation & Access Safeguards [5 Points]:** Neutron Security Groups/iptables/eBPF rules blocking inter-sandbox traffic and protecting the gateway management layer from sandbox access.

Required Deliverables :

- Benchmark report covering provisioning latency, gateway failover time, IP recycling speed, and load test metrics.
- Comprehensive API integration and security architecture documentation.

Phase 3: UI/UX & Documentation

Total score: 10 points

Required Features:

- **Administrator Management Dashboard [10 Points] :** Intuitive admin web dashboard displaying:

- Active team sandboxes & assigned OpenStack private IPs
- Real-time gateway health status (Primary vs. Standby)
- Mapped subdomains & live traffic logs
- Manual lifecycle controls (spin-up, teardown, manual gateway failover, and manual config rollback)

Required Deliverables :

- Administrative dashboard interface and frontend source code
- Technical report documenting the project, implementation, and results
- A demonstration video showcasing the project's features and functionality

Note:

- **Only the selected teams will advance to the Final Pitching phase.**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

- Present the completed solution to FelCloud technical representatives and the jury committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch
- A live demonstration (including simulated primary gateway crash & auto-failover on OpenStack)
- Q&A session with the jury members

Optional Advanced Features

Bonus Points

- **Automated SSL/TLS [+2 Points]:** Auto-issuance of wildcard or Let's Encrypt certificates for team subdomains.
- **Rate Limiting & Bandwidth Throttling [+2 Points]:** Gateway-level request capping per team.
- **Custom Domain Binding [+2 Points]:** Allow teams to map custom external domains to sandboxes.
- **Smart Health Monitoring & Fallback [+2 Points]:** Graceful maintenance/offline page when a sandbox crashes.
- **OpenStack Native Orchestration [+2 Points]:** Integration with OpenStack Heat, Designate (DNS), or eBPF/XDP filtering.

4. Recommended Skills

While you are highly encouraged to innovate and bring your own ideas to the table, your team will have a strong head start if you are familiar with:

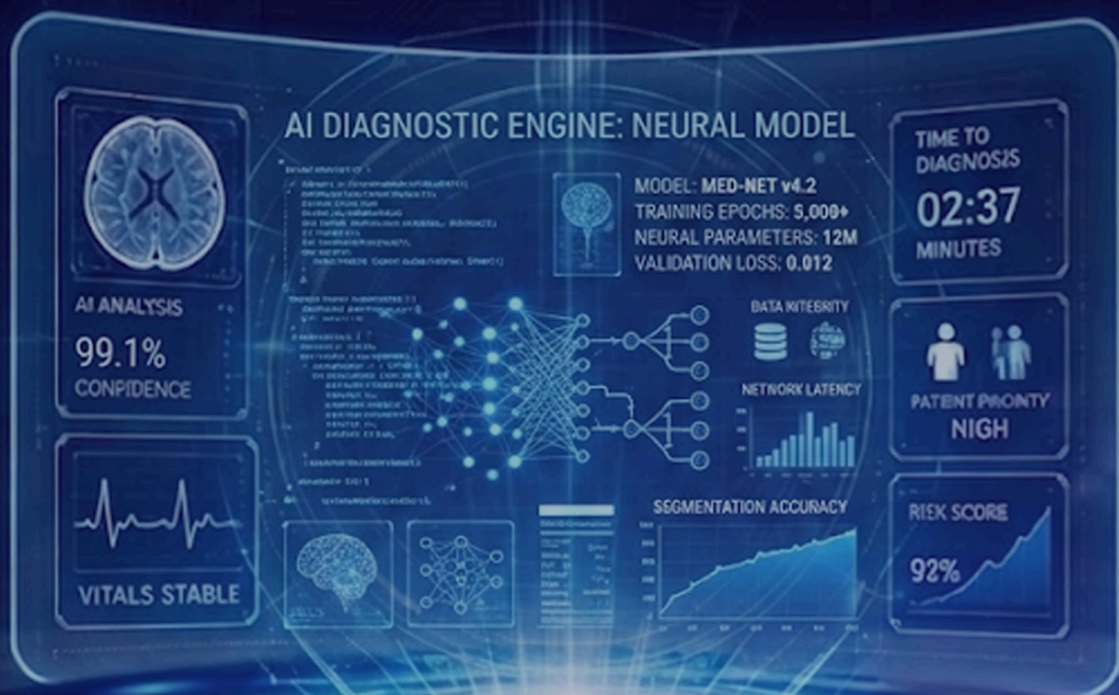
- **Cloud & Infra:** OpenStack (Nova, Neutron, Keystone, Octavia, Heat), KVM, Docker, Podman
- **Gateways & Routing:** Nginx, Traefik, HAProxy, Keepalived (VRRP), Open vSwitch, BGP
- **Backend & API:** Python (FastAPI/Flask), Go, Node.js, Rust
- **Frontend:** React, Vue.js, TailwindCSS
- **Networking:** OpenStack Neutron Security Groups, CoreDNS/Designate, iptables/eBPF

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- Awards
 - **First Place Team:** PFE 2027 opportunities at FelCloud
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Only the selected teams will advance to the Final Pitching phase.**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**



HealthTech

Challenge Title: ClaimGuard AI: Trustworthy Agentic Copilot for Healthcare Claim Pre-Validation

Challenge Reference: CSTAM-VELODOC

Company Collaborator: Velodoc Amazit

Assigned Mentor:

- Dr. Wael Hilali - Velodoc CTO
- Mr. Bilel Said - Velodoc CEO

1. Company Introduction & Challenge Context

Velodoc (by Amazit – FZCO) operates in the HealthTech & AI Solutions sector, specializing in AI-driven healthcare operational automation, claims workflow simplification, and administrative intelligence. To drive digital transformation and address emerging challenges, the company continuously explores innovations in Agentic AI & LLM Systems, Healthcare Interoperability (HL7 FHIR R4), and Cybersecurity, Privacy & Auditability.

In today's rapidly evolving landscape, where human-in-the-loop AI automation and real-time administrative validation is increasingly critical, we currently rely on manual claim review processes and rigid, traditional rule scripts. However, as our operations scale, we are actively seeking innovative approaches to automate pre-validation of healthcare claim packages, drastically reduce rejections and processing delays, and maintain total transparency, privacy, and auditability.

2. Problem Statement

Our main challenge is that healthcare claims are frequently delayed or rejected because administrative information is incomplete, inconsistent, duplicated, or unsupported at the point of submission. When high-volume synthetic claim packages with complex coverage rules, missing attachments, or data anomalies are submitted, our systems experience administrative bottlenecks, operational delays, and high denial rates. We require a robust solution capable of automatically and securely handling claim pre-validation, rule-linked evidence explanation, corrective action recommendations, and human escalation routing without disrupting existing administrative infrastructure or making unauthorized clinical decisions.

3. Technical Mission

Challenge Objective:

To design, prototype, and demonstrate a human-supervised Agentic AI Copilot system that ingests synthetic healthcare claim packages, detects administrative risks and policy inconsistencies, generates human-understandable explanations and recommendations, and logs all actions for auditability.

Participating teams must build a system evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Data Ingestion & Normalization [15 Points]:** Ingest synthetic claim packages (FHIR R4 JSON / CSV) containing patient, encounter, coverage, provider, diagnosis, and claim line items, and normalize them into a consistent internal representation.
- **Deterministic & AI Rule Engine [15 Points]:** Implement an automated validation engine executing against the fictional payer-rule catalogue (10–15 rules) to detect missing, inconsistent, duplicate, or unsupported data.
- **Explainability & Structured Output [10 Points]:** Generate structured, machine-readable validation findings containing Claim ID, Rule ID, rule-linked evidence, severity level, confidence score, and suggested corrective action.
- **Audit Log Engine [10 Points]:** Systematically record an auditable, immutable log of all checks, AI recommendations, confidence scores, and system decisions.

Required Deliverables :

- Source Architecture Diagram & Data Flow Documentation.
- A short video demonstration of the MVP.
- Code repository with clear installation and execution instructions.

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Detection Quality & Benchmark Performance [15 Points]:** Evaluate system performance on the validation dataset (50 claims) achieving high Macro F1 score across rule categories and preserving valid claims (low false-positive rate).

- **Human-in-the-Loop & Escalation Logic [10 Points]:** Implement reviewer-facing routing that flags low-confidence or high-severity cases for explicit human approval, allowing manual overrides and feedback logging.
- **Privacy, Security & Safety Guards [5 Points]:** Implement data minimization, access control, prompt /data security guards (preventing hallucinated clinical advice), and graceful error handling for malformed FHIR inputs.

Required Deliverables :

- Test evaluation report (detection metrics, F1 score, false positive rate, latency, and limitations).
- Privacy and Security note (threat model, access control, least privilege, and auditability).

Phase 3: UI/UX & Documentation

Total score: 10 Points

Refine the project for final evaluation by improving usability and preparing all required deliverables.

Required Features:

- **Quality of Interface/Usability [10 Points]:** An intuitive human-review interface (Web UI, dashboard, interactive notebook, or API GUI) presenting findings, rule evidence, severity levels, and interactive approval/override controls.

Required Deliverables :

- Finalized UI/UX design and user experience improvements.

- Technical report documenting the project, implementation, and results.
- A demonstration video showcasing the project's features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

- Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional pitch deck supporting the presentation.
- A live demonstration of the Agentic AI Copilot interface and core features.
- Q&A session with the jury members.

Optional Advanced Features

Bonus Points

- Multi-Format Attachment Parsing (OCR/RAG for PDF clinical notes) **[2 Points]**
- Dynamic Payer-Rule Management GUI/API **[2 Points]**
- Active Learning Feedback Loop from Human Overrides **[2 Points]**
- Cryptographic / Local Ledger Audit Log Security **[2 Points]**
- Real-Time Async Streaming REST / WebSocket API **[2 Points]**

4. Recommended Skills

While you are highly encouraged to innovate and bring your own ideas to the table, your team will have a strong head start if you are familiar with:

- **Software & Frameworks:** Python, LangChain / LangGraph, AutoGen / CrewAI, FastAPI / Flask, React / Next.js, Docker
- **Protocols & Standards:** REST APIs, WebSockets, JSON/JSON-LD, FHIR / HL7 (Optional)
- **AI & Security:** Prompt Engineering, Retrieval-Augmented Generation (RAG), PII Anonymization, Role-Based Access Control (RBAC)

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Remote Summer internships and PFE 2027 opportunities at Velodoc
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all challenges**

Automation



Challenge Title: Automated RFP Intelligence & Commercial Proposal Generation System

Challenge Reference: CSTAM-OliveSoft

Company Collaborator: OliveSoft

Assigned Mentor:

- Khalil Masmoudi - OliveSoft AI & Data Engineer

- Bilel Djemel - OliveSoft AI & Data Engineer

1. Company Introduction & Challenge Context

OliveSoft operates in the IT Services & Software Engineering sector, specializing in enterprise digital solutions, web/mobile engineering, and custom software development. To drive digital transformation and address emerging business challenges, the company continuously explores innovations in Generative AI & LLMs, Multi-Agent Orchestration (n8n), and Retrieval-Augmented Generation (RAG).

In today's rapidly evolving software market, where rapid response times and tailored commercial proposals for public/private RFPs (Appels d'Offres) are increasingly critical, we currently rely on manual market monitoring, fragmented internal archives (past projects, client history, CV databases, tech stacks), and manually formatted commercial proposals. However, as our operations scale, we are actively seeking innovative approaches to automate tender discovery, conduct AI-driven prospect research, and generate customized commercial proposal presentations.

2. Problem Statement

Our main challenge is the time-intensive, manual process of identifying relevant RFPs, gathering prospect intelligence, aligning internal expertise (CVs, past projects, tools), and drafting comprehensive commercial proposals.

When a new tender or RFP is published or submitted, our teams experience delays in evaluation, high manual research overhead, and risk of suboptimal positioning due to fragmented access to internal knowledge assets. We require a robust solution capable of automatically detecting market leads, conducting agentic prospect research, retrieving matching OliveSoft capabilities via RAG, and generating customized commercial presentations and proposals without disrupting existing workflows.

3. Technical Mission

Challenge Objective:

To design, prototype, and demonstrate an end-to-end AI-powered system that monitors market tenders, enriches context through autonomous prospect research, leverages OliveSoft's internal knowledge base via RAG, and automatically generates tailored commercial proposal presentations using n8n orchestration and Generative AI.

Participating teams must build a system evaluated across the following core phases:

Phase 1: Core Functionalities MVP (Mandatory)

Total score: 50 points

Required Features:

- **Lead & Tender Detection Engine [15 Points]:** Automated ingestion and detection of published tenders/RFPs from public datasets, simulated feeds, or structured inputs.
- **Robust RAG Knowledge Retrieval [15 Points]:** A vector-search pipeline capable of indexing and extracting knowledge from Olivesoft's internal documentation (past project references, client portfolios, CV bank, tools, and technical stacks).
- **Agentic Prospect Research [10 Points]:** Autonomous AI agent workflow to research prospect details (sector, estimated revenue, past projects, key partners, specific domain requirements).
- **Workflow Orchestration with n8n [10 Points]:** Seamless orchestration using n8n to connect detection, research, RAG retrieval, and initial document generation.

Required Deliverables :

- System architecture diagram and data pipeline flow documentation
- Source code repository (including n8n workflows and RAG implementation)
- A short video demonstration of the MVP

Phase 2: Integration & Testing

Total score: 30 Points

Required Features:

- **Generative Presentation & Proposal Output [15 Points]:** Automated drafting of commercial proposal presentations matching Olivesoft's commercial templates and branding guidelines.
- **RAG Benchmark & Retrieval Precision [10 Points]:** Benchmarking and optimizing semantic search accuracy to ensure correct CV and project matching based on tender requirements.

- **Edge Case & Data Validation Handling [5 Points]:** Graceful handling of incomplete tender specifications, noisy web research data, or missing internal references.

Required Deliverables :

- RAG benchmark, evaluation report, and test suite execution results.
- Detailed integration & API workflow documentation.

Phase 3: UI/UX & Documentation

Total score: 10 Points

Refine the project for final evaluation by improving usability and preparing all required deliverables.

Required Features:

- **Quality of Interface / Usability [10 Points]:** Intuitive web interface/dashboard allowing sales managers to view detected leads, inspect matched internal assets, review prospect research, edit generated parameters, and download proposal decks.

Required Deliverables :

- Finalized UI/UX design and user experience dashboard.
- Technical report documenting project architecture, LLM/RAG strategies, and results.
- A demonstration video showcasing full end-to-end platform features and functionality.

Note:

- **Only the selected teams will advance to the Final Pitching phase**
- **The detailed selection criteria are provided in the General CSTAM 3.0 Challenge Guide**

Phase 4: Pitching

Total score: 10 Points

Present the completed solution to the company representatives and the evaluation committee during the CSTAM 3.0 event.

Required Deliverables :

- A professional presentation supporting the pitch.
- A live demonstration of the platform.
- Q&A session with the jury members.

Optional Advanced Features

Bonus Points

- **Multi-Format Deck Exporting [+2 Points]:** Direct generation and export to editable .pptx slides and formatted .pdf documents.
- **Automated Requirements Matrix & Score [+2 Points]:** Auto-generating a matrix scoring OliveSoft's technical coverage against RFP requirements.
- **Competitor & Budget Insights Agent [+2 Points]:** Agentic estimation of prospective budget ranges and competitive positioning analysis.
- **Human-in-the-Loop AI Refinement [+2 Points]:** An interactive UI feature allowing sales teams to prompt-edit specific slide sections in real-time.
- **Multi-Source Crawler Integration [+2 Points]:** Ingestion pipelines for real-time crawling across multiple public procurement portals.

4. Recommended Skills

You're encouraged to bring your own ideas, but familiarity with the following helps:

- **Hardware / Infrastructure:** Cloud
- Services, Docker Containers, Serverless Deployments.
- **Software & Frameworks:** n8n Workflow Automation, Python, LangChain / LlamaIndex, Fast API, React / Next.js / Tailwind CSS, Vector Databases (Qdrant, ChromaDB, Pinecone, or pgvector).
- **Protocols & Models:** REST APIs, Webhooks, OpenAI / Anthropic APIs, Open-Source LLMs (Ollama / vLLM), JSON-Schema formatting.

5. Rewards & Opportunities

- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Summer internships and PFE 2027 opportunities at OliveSoft
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- **Additional rewards may be announced later**
- **Teams will also compete for the Global CSTAM Book Grand Prizes Top 3 across all challenges**

General CSTAM Book Guidelines

While each team will tackle a unique industrial problem proposed by our collaborator partner companies, the core rules, expectations, deliverables, and support frameworks remain consistent across all projects.

1. Team Eligibility & Structure

Team Requirements:

- Open to all students.
- Eligible for IEEE members and Non IEEE Members.
- Each participant may join **only one team**.
- Each team may apply for **only one technical challenge**.

Team Composition:

- Teams must consist of **3 to 5 members**.
- **Bonus Points:** Teams composed of members from **different levels of study** will receive a **bonus of 5 points** during the evaluation process.
- Interdisciplinary teams are strongly encouraged.

2. CSTAM Book Timeline

2.1. Initial Registration:

Registration Opens: August 20 , 2026

Teams must apply through the official CSTAM Book Registration Form, where they are required to :

- Provide the team's general information.
- Rank their preferred technical challenges.
- Provide their related information to preferred project
- Accept and sign the **CSTAM Book Participant Commitment**, acknowledging the rules, responsibilities, and terms governing the CSTAM Book.

2.2. Infosessions:

Prior to the project selection phase, each partner company will host an information session to present its technical challenge and answer participants' questions, helping them make informed decisions when ranking their project preferences.

2.3. Challenge Selection Phase:

Acceptance into a specific company challenge will be evaluated based on:

- Participant CVs and academic backgrounds.
- A brief statement of motivation.
- Evidence of previous similar projects or relevant technical experience.

Note: Accepted teams will be notified via email.

2.4. Projects Development Phase (Phase 1 & 2):

Launching Date: Upon announcement of the selected teams

The challenge follows a strictly phased workflow to ensure steady progress. Throughout the project development phase, teams must actively develop their solutions, meet the objectives and deliverables of each phase, and adhere to the established timeline.

The following schedule defines the key phases and deadlines:

Phase 1 : Core Functionalities MVP (Mandatory)

Total score: 50 points

Deadline: October 1, 2026

Teams are expected to deliver the initial version of their solution, including the core functionalities and proposed technical approach.

Phase 2 : Integration & Testing

Total score: 30 points

Deadline: October 20, 2026

Teams will focus on integrating project components, improving performance, and validating the functionality of their solution.

Phase 3 : UI/UX & Documentation (Selection Phase)

Total score: 10 points

Deadline: November 1, 2026

Only selected teams will advance to the Pitching phase and selection criterias will be based on:

- Source code repository
- Proof of Concept (PoC)
- **Technical Report Requirements:**
 - Abstract
 - Team Overview & Contributions
 - Problem Statement & Industrial Context
 - Proposed Solution
 - System design
 - Mockups
 - Technologies, Frameworks, and Tools Used
 - Development Milestones
 - Testing results
 - Future Improvements
 - References (if applicable)
- Commitment to the Phases required deliverables
- Engagement and commitment in mentoring sessions
- **Late Submission Penalty:** 5 Points will be deducted for the late submission of required deliverables

Note: Accepted teams will be notified via email.

Phase 4 : Pitching

Total score: 10 points

Deadline: In Person at CSTAM 3.0

Selected teams will present their solutions before the company representatives and the evaluation committee through a final pitch and live demonstration during the CSTAM 3.0 event.

The following guidelines apply to the pitching sessions:

- **Language Requirement:** All presentation, Technical Report, demonstration, and Q&A sessions must be conducted strictly in **English**.
- **Team Representation:** A maximum of **two (2)** team members may participate in the final pitching phase.
- **Format & Time Allocation for the Presentation (12 Minutes in Total):**
 - **5 minutes:** Presentation
 - **2 minutes:** Live Demonstration
 - **5 minutes:** Q&A Session

Optional Bonus

- **Total score: Up to 12 points**

Rewards exceptional creativity, "outside-the-box" engineering, or advanced stretch goals. Judges may award bonus points for complex, secondary features that exceed basic MVP requirements to significantly elevate the project.

Winners Announcement

- **In Person at CSTAM 3.0**
- **Number of Winners:** 3 teams
- **Awards**
 - **First Place Team:** Summer internships and PFE 2027 opportunities at Company Challenge Owner
 - **Second Place Team:** Certificate of Recognition
 - **Third Place Team:** Certificate of Recognition

Note:

- Additional rewards may be announced later
- Teams will also compete for the Global CSTAM Book Awards Top 3 across all subjects

3. Global CSTAM Book Awards :

The scores from all company challenges will be centralized to establish the overall CSTAM Book ranking.

The highest scoring teams will be awarded the 1st, 2nd, and 3rd Place Overall CSTAM Book Titles based on the following criteria:

- **Overall Project Score:** Final score obtained from all challenge phases.
- **Technical Excellence:** Quality of implementation, architecture, and technical solution.
- **Innovation & Impact:** Level of originality and added value.
- **Project Completion & Deliverables Quality:** Compliance with requirements, documentation quality, and maturity of the final solution.
- **Presentation & Demonstration Performance:** Quality of the final pitch, clarity of communication, and effectiveness of the live demonstration.
- **Team Commitment & Collaboration:** Engagement, progress throughout the challenge, and ability to work effectively as a team.

4. AI Usage Policy :

The use of Artificial Intelligence (AI) tools is permitted throughout the development of the project, provided that the following guidelines are respected:

- Teams may use AI-assisted tools (e.g., ChatGPT, GitHub Copilot, Claude, Gemini) to support research, development, debugging, documentation, and design activities.
- All AI-generated code, content, or design artifacts **must be carefully reviewed, validated, and understood by the team before submission.**
- Teams are fully responsible for the accuracy, originality, security, and quality of all submitted deliverables, regardless of the tools used during development.
- During the final presentation and Q&A session, participants must be able to clearly explain and justify any AI-assisted contributions.

Note: The use of AI tools does not replace the team's technical understanding or expertise.

5. Mentoring Sessions & Technical Support

Throughout the project development journey, each team will benefit from guidance provided by a dedicated technical mentor from the corresponding company partner.

Mentoring sessions will combine **workshop sessions and project supervision.**

Role of Mentors:

The mentor will support teams by:

- Providing clarification on project objectives, requirements, and expected deliverables.
- Guiding teams in defining appropriate technical approaches and solution architectures.
- Following project progress, providing feedback, and helping resolve technical challenges.
- Supporting teams during implementation, integration, testing, and validation stages.
- Sharing technical insights and best practices to ensure alignment with phases requirements.

Role of Teams:

- Mentoring sessions are **mandatory for all teams** throughout the project development phases.
- Each team must ensure that its **Team Leader** represents the team during the supervision sessions.
- The Team Leader will be responsible for sharing the team's progress, challenges and questions with the mentor during the supervision session.
- Mentoring session attendance, participation, and engagement will be taken into consideration as part of the final evaluation criteria

Note: Teams will be notified via email about their mentoring sessions.

6. Participant Commitment

By registering and proceeding with a CSTAM Book challenge, all team members must agree to the **CSTAM Book Participant Commitment Agreement.**

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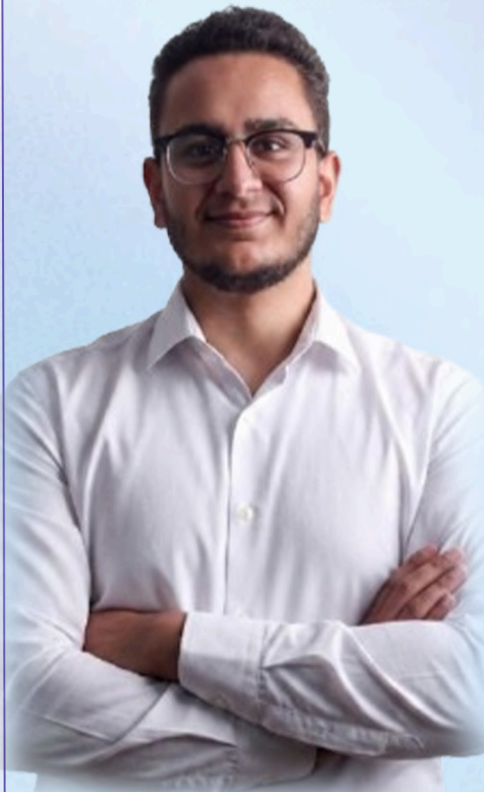


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